Understanding convolution

From now on we will define any system as $h(n)$, where $h(n) = \text{System}(\delta[n])$

Before understanding the how the system physically applies convolution lets try out one ourselves, just learn how to use the formula.

Let $x[n] = \{1, \underset{\uparrow}{2}, 3\}$

$$h[n] = \{\underset{\uparrow}{-1}, 1, 2\}$$

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

x is only non-zero at $k = -1, 0, 1$

So, $y[n] = x[-1] \cdot h[n+1] + x[0] \cdot h[n] + x[1] \cdot h[n-1]$

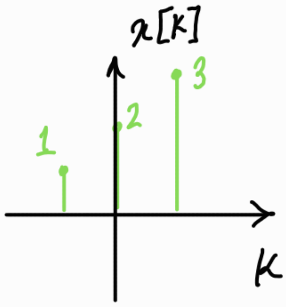
$$= 1 \cdot \{-1, \underset{\uparrow}{1}, 2\} + 2 \cdot \{\underset{\uparrow}{-1}, 1, 2\} + 3 \cdot \{\underset{\uparrow}{0}, -1, 1, 2\}$$

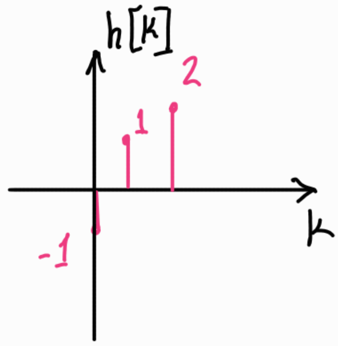
$$= \{-1, \underset{\uparrow}{1}, 2\} + \{\underset{\uparrow}{-2}, 2, 4\} + \{\underset{\uparrow}{0}, -3, 3, 6\} = \{-1, \underset{\uparrow}{-1}, 1, 7, 6\}$$

Now we have to understand what the formula is doing graphically.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

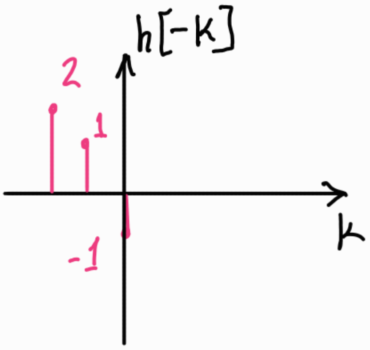
At first we are replacing the n-axis with k axis. Let work the same example but we will go through the process step by step.



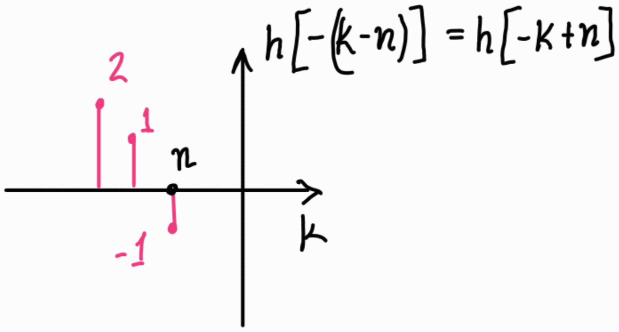


Changing the axis.

Now we find out $h[-k]$. This reversal process is a key feature of convolution.

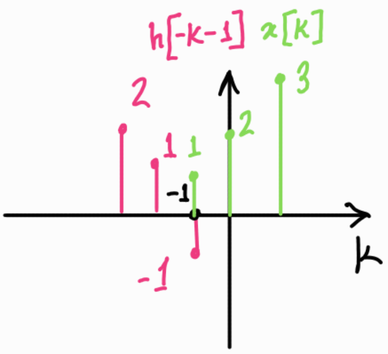


Then we shift the signal by n. Remember to shift by the 0 point in case you are confused, even if there is no signal there.



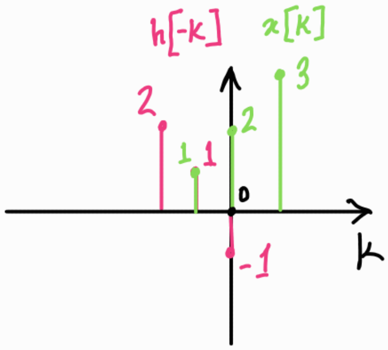
Now you take this new signal, change the n value. Multiply the signals and get $y[n]$.

@ $n = -1$



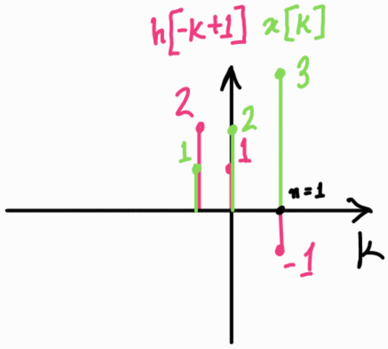
$y[-1] = -1$

@ $n = 0$



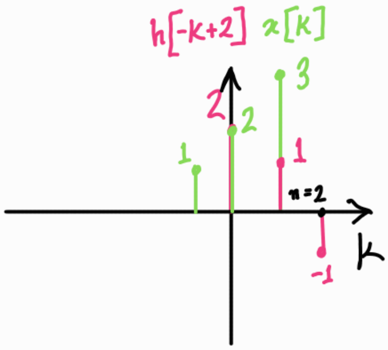
$y[-1] = -1$

@ $n = 1$



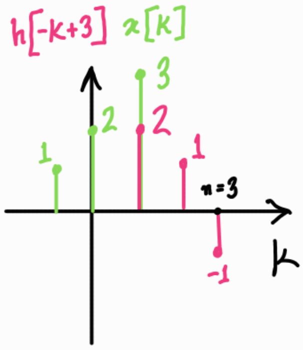
$y[1] = 1$

@ $n = 2$



$y[2] = 7$

@ $n = 2$



$y[3] = 6$

So, $y[n] = \{-1, \underset{\uparrow}{-1}, 1, 7, 6\}$ ⟵ same as before

Note — Lowest output index = lowest x index + lowest h index

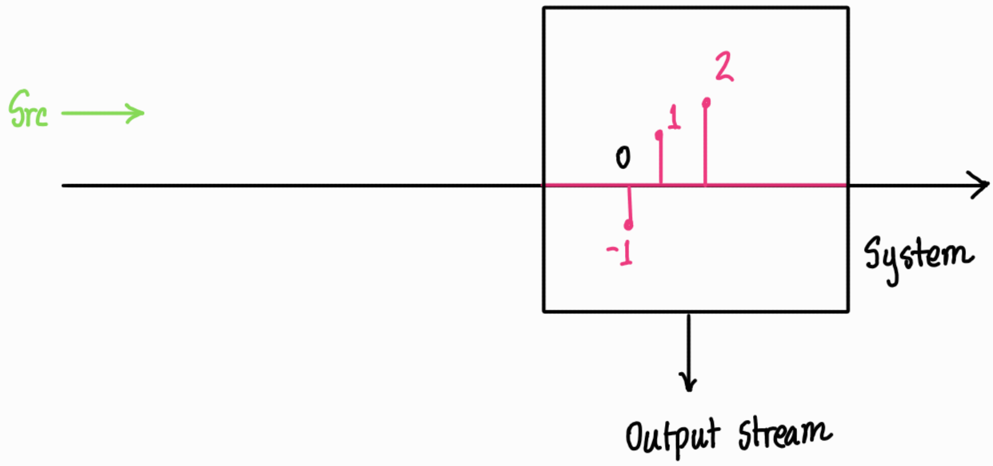
Highest output index = highest x index + highest h index

Note — Convolution is commutative. So, $x[n] * h[n] = h[n] * x[n]$

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot x[n-k]$$

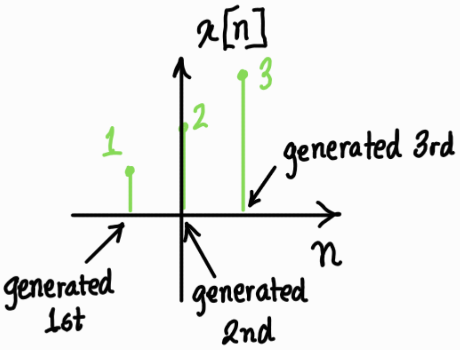
So, $x[n]$ can also be reversed and it is more natural to be honest.

Lets see why this reversal is important.

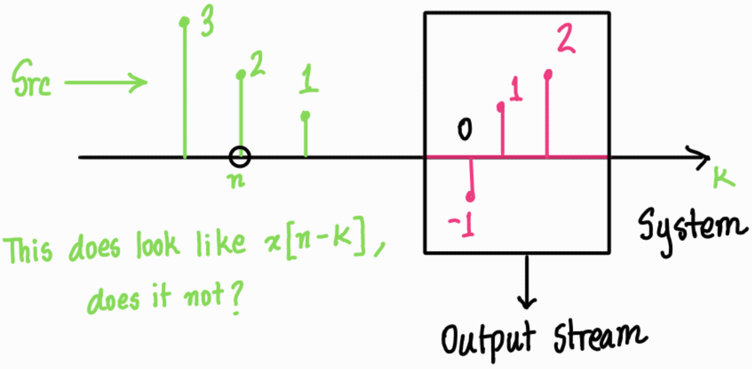


This is the general system. We will assume that the signal travels as a stream from the source to the system one pulse at a time, and then out of the system through the other side.

Lets take a look at $x[n]$



So the pulse stream from the source will look like —



This does look like $x[n-k]$, does it not?

This is why we have to reverse the signal because natural signals are generated one pulse at a time.

You can work out this system and you will reach the same conclusion.

Note — We will generally reverse $h[n]$ instead of $x[n]$.

Question — What if we didn't reverse either of the signals?

That operation is called a cross-correlation operation.

$$x[n] \star h[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[k-n]$$

note that unlike convolution, $x[n] \star h[n] \neq h[n] \star x[n]$

Cross correlation is done to find similarity between signals.

$x[n] \star h[n]$ means you will slide $h[n]$ over $x[n]$ to find how similar $h[n]$ is compared to $x[n]$.

We will come back to cross-correlation once we move to convolutional neural networks.

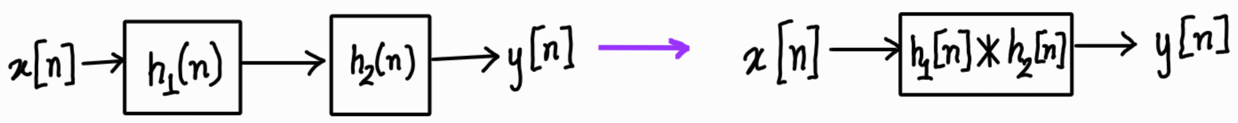
Properties of convolution & system thinking —

We will just learn 3 properties and observe their implications on systems.

(1) Convolution is associative —

$$\left(x[n] * h_1[n]\right) * h_2[n] = x[n] * \left(h_1[n] * h_2[n]\right)$$

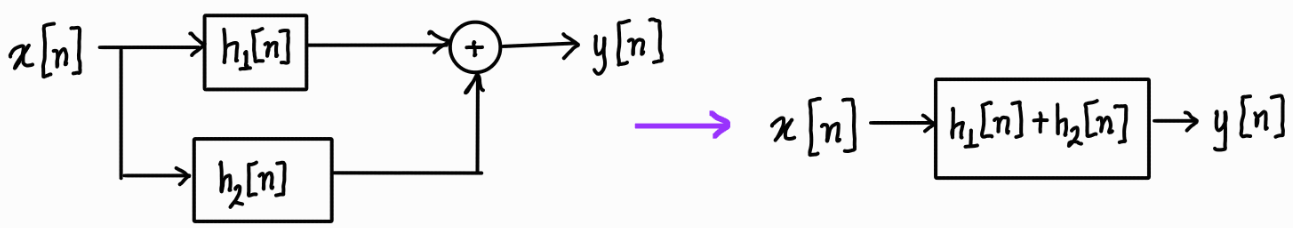
In terms of systems,



(2) Convolution is distributive —

$$x[n] * \left(h_1[n] + h_2[n]\right) = x[n] * h_1[n] + x[n] * h_2[n]$$

In terms of systems,



③ Convolution with $\delta[n-k]$ —

$$x[n] * A\delta[n-k] = Ax[n-k]$$

when, $k=0$ — $x[n] * A\delta[n] = Ax[n]$

So sometimes it maybe wise to break down smaller signals into impulse functions.